

TABLE OF CONTENTS

TABLE OF CONTENTS	1
1- OVERVIEW	2
2- INPUT/OUTPUT CONNECTIONS	4
3- COMMAND DEFINITIONS	10
3-1 Low Power Operate State	10
3-2 Error Data	11
3-2-1 Error/Status Reporting	11
REVISION HISTORY	14

1- OVERVIEW

The SRFD II RF Amplifier and Interface (SRFD II Module) is the second phase in the development of the scaleable RF Driver (SRFD) subsystem. This component is a combination of the RF Interface (RFI) and the RF Power Module (RFM) as defined in the original SRFD architecture. The SRFD II Module interfaces with the System Support Module (SSM), the Transceiver Exciter Output, and the RF coils through the Heliax cable interface. The SRFD II Module performs the following functions: Head/Body mode switching, serial link decoding, control and status communication between the system and the RF amplifier, RF power amplification, and RF power sampling for power monitoring. The SRFD II Module provides appropriate test points, displays and adjustments to allow for calibration and troubleshooting in the field.

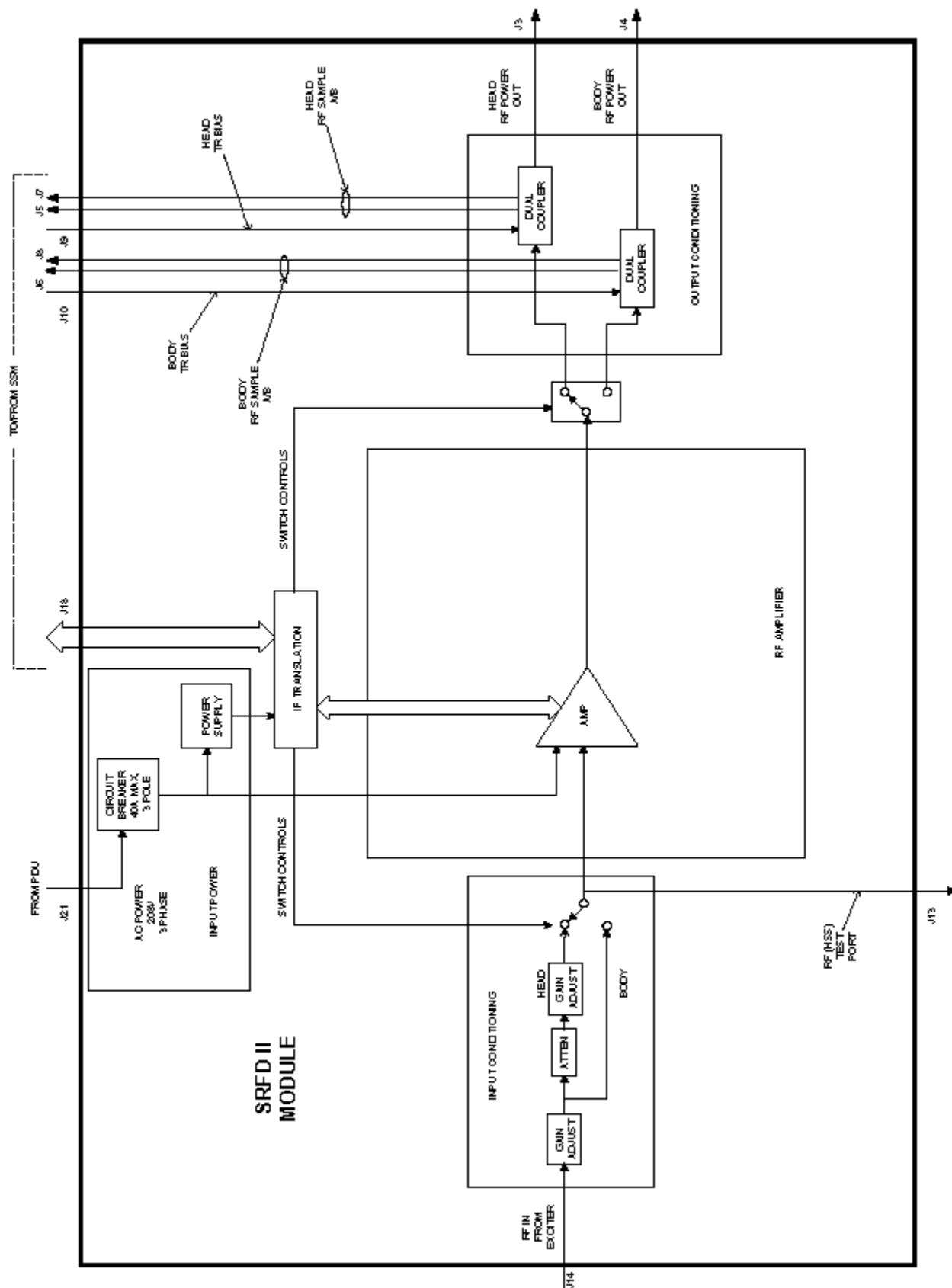
The SRFD II Module consists of five functional blocks: IF Translation, Input Conditioning, RF Amplifier, Output Conditioning and Input Power, as shown in the block diagram of Illustration 1-1. The IF Translation block converts the commands from the Signa Data Acquisition system (coming through the SSM) into the appropriate RF Amplifier control signals. It provides status information about the module from the SRFD II back to the Data Acquisition system, and to the user via a front panel display of LED's or an LCD. It also provides the internal control signals for Head/Body mode switching within the SRFD II.

The Input Conditioning block allows for the independent adjustment of the Head and Body RF input levels. It switches the input to the RF Amplifier block to the mode (Head or Body) required by the scan, and provides an RF Test Port for calibration and field service testing purposes with the High Speed Stability (HSS) field service tool.

The RF Amplifier block provides the gain necessary to produce the RF transmit pulses needed for MR imaging. If needed for normal installation or field service maintenance, adjustments and test ports are to be provided to allow for optimization of the amplifier module parameters; however, no adjustments are required by this specification.

The Output Conditioning block contains the Dual Directional Couplers that provide the redundant RF Samples A/B in Head/Body Mode to the Power Monitor. It also completes the Head/Body switching function by routing the RF power to the appropriate output port. The Head and Body TR Bias signals are to be coupled into the RF output lines within this block. The Head and Body TR Bias lines are to be electrically isolated from ground and from each other. The total DC path loss from the TR Bias inputs to the respective Head and Body RF Ports is to be very small and controlled.

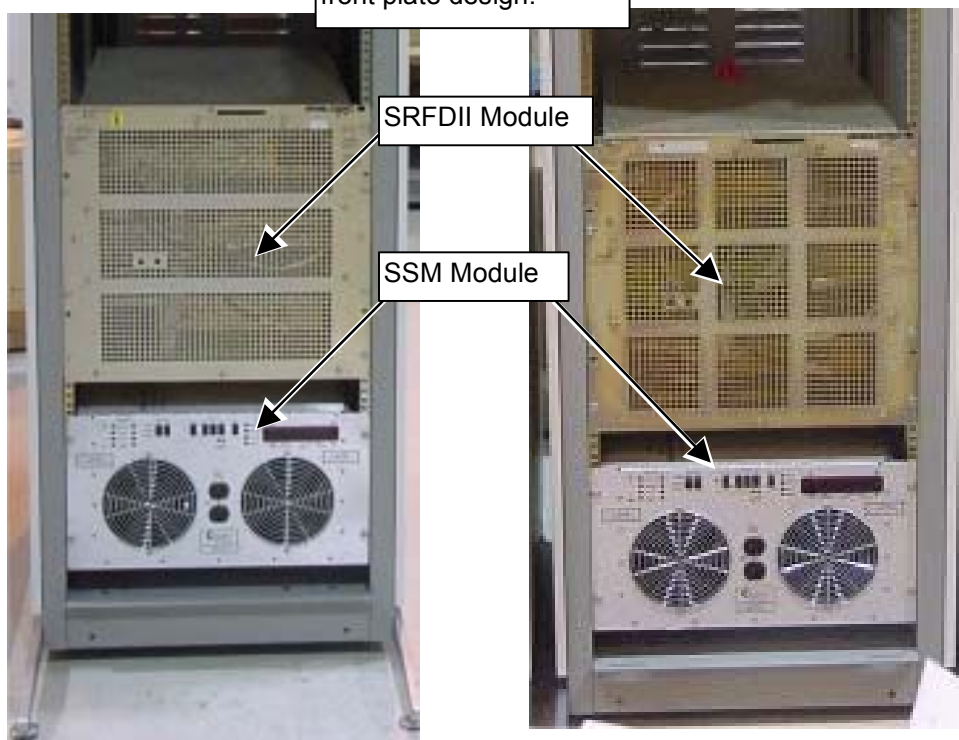
The Input Power block provides the line interface for the SRFD II Module. AC power to the module is supplied by the system Power Distribution Unit (PDU). The Input Power block provides a rear panel power off circuit breaker, with a front panel display to indicate AC power on/off. It also contains any power supplies necessary to provide regulated power to the low power circuitry in the module.



SRFD II BLOCK DIAGRAM
ILLUSTRATION 1-1

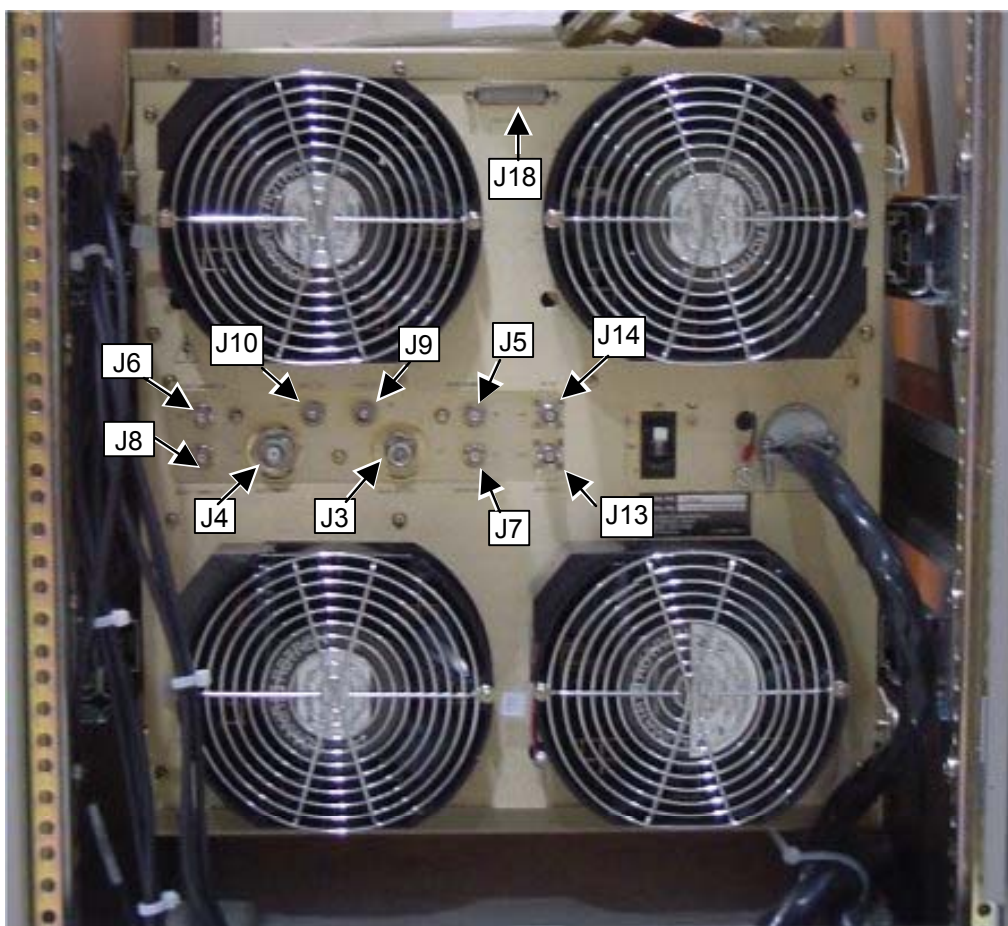
2- INPUT/OUTPUT CONNECTIONS

Note: The early SRFDII modules have a different front plate design.



SRFD2 FRONT VIEW
ILLUSTRATION 2-1

2- INPUT/OUTPUT CONNECTIONS (Continued)



SRFD2 REAR VIEW
ILLUSTRATION 2-2

2- INPUT/OUTPUT CONNECTIONS (Continued)

TABLE 2-1
INPUT CONNECTORS

Designator	Chassis Label	Description	Connector Type	Signal Type	Signal Specification
J9	HEAD TR	Head TR Bias Input	BNC Female	Switching DC control	± 15 VDC, < 2 A rms; DC resistance to J3 center pin < 0.2 ohm; electrically isolated from ground and from Body TR
J10	BODY TR	Body TR Bias Input	BNC Female	Switching DC control	± 15 VDC, < 2 A rms; DC resistance to J4 center pin < 0.2 ohm; electrically isolated from ground and from Head TR
J14	RF IN	RF Input	BNC Female	Low level RF	-4 to +10 dBm, depending on system exciter configuration; SRFDII shall produce rated output power for any RF Input Level in this range
J18	COMM	General Communications	25 pin Sub-D Male RS 232/422	TTL	See Table 2-2
J21	LINE IN	Line Power In	10ft power pigtail	3 phase line power	208 VAC $\pm 10\%$ nominal, 3 phase, 40 A maximum, 50/60 Hz ± 3 Hz. Normal system connection shall be to isolated side of system PDU transformer.

2- INPUT/OUTPUT CONNECTIONS (Continued)

TABLE 2-2
J18 COMM CONNECTION PIN SIGNALS

Signal Name Conventions: “/” = active low signal

“P” = positive signal connection

“N” = negative signal connection

“NC” = no signal connection through system I/F cable

“NOT USED” = signal connection present but not used by Signa for amp control

J18 Pin #	Signal Name	Signal Type	Description
1	RFLCK-N	RS422	PWR MON 1 fault from SSM; inhibit UNBLANK in AMP
14	RFLCK-P	RS422	PWR MON 1 fault from SSM; inhibit UNBLANK in AMP
2	NC	---	---
15	/RFLCK-P	RS422	PWR MON 2 fault from SSM; inhibit UNBLANK in AMP
3	/RFLCK-N	RS422	PWR MON 2 fault from SSM; inhibit UNBLANK in AMP
16	/UNBLANK-N	RS422	UNBLANK echo from AMP to SSM
4	/UNBLANK-P	RS422	UNBLANK echo from AMP to SSM
17	NC	---	---
5	/HVREL-EN	TTL Pullup (1)	PWR MON 1 or 2 fault from SSM; inhibit UNBLANK in AMP
18	RXD	RS232	Serial Communications, data from SSM
6	NOT USED	---	---
19	GND	---	---
7	NOT USED	---	---
20	NOT USED	---	---
8	NOT USED	---	---
21	GND	---	---
9	/UNBLANK-P	RS422	UNBLANK input to AMP from SSM
22	/UNBLANK-N	RS422	UNBLANK input to AMP from SSM
10	TXD	RS232	Serial Communications, data to SSM

2- INPUT/OUTPUT CONNECTIONS (Continued)

TABLE 2-2 (Continued)
J18 COMM CONNECTION PIN SIGNALS

J18 Pin #	Signal Name	Signal Type	Description
23	NC	---	---
11	NC	---	---
24	NC	---	---
12	NC	---	---
25	NC	---	---
13	NC	---	---

(1) Pullup through 10kOhm to +5V.

2- INPUT/OUTPUT CONNECTIONS (Continued)

TABLE 2-4
OUTPUT CONNECTORS

Designator	Chassis Label	Description	Connector Type	Signal Type	Signal Specification
J3	HEAD RF OUT	Head RF Power Out	N Female	High Power RF	2 kW PEP
J4	BODY RF OUT	Body RF Power Out	HN Female	High Power RF	16 kW PEP
J5	HEAD SAMPLE B	Head RF Sample B	BNC Female	Low level RF	-40 dB $\pm$ 0.1 dB, directivity > 15 dB; rear panel adjustable if necessary
J6	BODY SAMPLE B	Body RF Sample B	BNC Female	Low level RF	-50 dB $\pm$ 0.1 dB, directivity > 15 dB; rear panel adjustable if necessary
J7	HEAD SAMPLE A	Head RF Sample A	BNC Female	Low level RF	-40 dB $\pm$ 0.1 dB, directivity > 15 dB; rear panel adjustable if necessary
J8	BODY SAMPLE A	Body RF Sample A	BNC Female	Low level RF	-50 dB $\pm$ 0.1 dB, directivity > 15 dB; rear panel adjustable if necessary
J13	HSS/TEST	RF (HSS) Test Port	BNC Female	Low level RF	400 mW $\pm$ 10% (26 $\pm$ 0.44 dBm) with peak RF Input level set for rated power output; HSS output shall be gated (blanked & unblanked) as the RF outputs are.

3- COMMAND DEFINITIONS

The Signa Data Acquisition System (DACQ) commands the SRFD II Module to one of three logical states through the I/F Translator. Only the following state transitions are valid and are commanded by the Signa system:

POWER OFF $\Rightarrow$ STANDBY

STANDBY $\Rightarrow$ OPERATE

OPERATE $\Rightarrow$ STANDBY

OPERATE $\Rightarrow$ POWER OFF

STANDBY $\Rightarrow$ POWER OFF

The physical states of the SRFD II corresponds to the expected logical states as follows:

AC LINE OFF $\equiv$ AC line circuit breaker in the OFF position; no function at any level in the module. This does not equate to one of the logical states.

POWER OFF $\equiv$ Low power dissipation state; control boards on, High Voltage Supplies not charged, amplifier gating (UNBLANK) inhibited, fans may be OFF or ON. Module is IDLE.

STANDBY $\equiv$ All supplies powered on; high voltage supplies charged, fans on, but amplifier gating is inhibited. Amp is READY to transmit RF.

OPERATE $\equiv$ All supplies powered on; amplifier gating enabled; module is amplifying the RF input.

The SRFD II is capable of operating in one of three modes: BODY, HEAD or TEST. The Signa DACQ commands the module to change modes only when the amplifier is in the STANDBY state. The default mode upon power up with the AC line circuit breaker is BODY. The transition time to change between modes is < 500 msec.

3-1 Low Power Operate State

The SRFD II Module is capable of operating in a special low output power state, OPERATE LOW, under certain device failure conditions. The exact failure conditions that can be bypassed, and the amount of output power available from the amplifier in this state is defined by Analogic. The Signa host software does not specifically command the amplifier into OPERATE LOW; rather, the amplifier is commanded to OPERATE as usual. OPERATE LOW is indicated to Signa upon reading the status register.

3-2 Error Data

The error register is read whenever the bit in the status register indicates there is a fault to report. If an error condition exists at the time of power up, the power up state is positive for those faults. The time between power turn on with the circuit breaker and status signals being valid is < 5 seconds.

In the event of an error, the Signa DACQ system commands the amplifier into POWER OFF state to reset the error if possible. If the error has been fixed, the error register is cleared. The DACQ will then command the amplifier back up to the state it was in when the error occurred. This cycle will repeat indefinitely until the error is fixed and the error bit does not indicate an error. In the event of multiple power monitor (PMON) errors, these errors are stacked in the SRFD2 host interface software and cleared one at a time via this command cycle.

Refer to Table 3-1 for a definition of PMON errors.

3-2-1 Error/Status Reporting

The SRFD II reports and indicates the error and status conditions of the module by 2 mechanisms: by a visual indication of high level information, and by reporting the exact error/status codes through the communications I/F.

Table 3-1 is a listing of the generic error and status conditions that the SRFD II module reports and visually indicates.

The visual indication is a visual display on the front panel of the module consisting of an array of LED's. The front panel display follows the TEXT and COLOR coding as defined in Table 3-1.

Reporting of the module error codes is done through the error register using the error codes listed in Table 3-1. It is not required that all error codes be utilized; the table shows the generic error conditions that the Signa DACQ software is defined to interpret. Any error code reported that is not defined in this table will be interpreted as an unknown error.

3-2-1 Error/Status Reporting (Continued)

TABLE 3-1
ERROR/STATUS DEFINITIONS

Error/Status	Condition Front Panel Display
POWER LOSS – Amplifier internal power loss condition; I/F still active	Set FAULT indication (RED)
POWER SUPPLY FAULT - Amplifier high voltage power supply failure	Set FAULT indication (RED)
DUTY CYCLE – Maximum UNBLANK duty cycle exceeded	Set FAULT indication (RED)
CURRENT OVERLOAD - Instantaneous peak current in RF device(s) exceeds spec	Set FAULT indication (RED)
DEVICE FAULT – RF device shorted	Set FAULT indication (RED)
HV LOW – Amplifier high voltage power supply voltage dropped below specified minimum	Set FAULT indication (RED)
HV HIGH – Amplifier high voltage power supply voltage went above specified limit	Set FAULT indication (RED)
DRIVE - RF Drive level to amp exceeds specification	Set FAULT indication (RED)
HEAT - RF Heat sink temperature exceeds specification	Set FAULT indication (RED)
PEAK POWER – Peak power from amplifier exceeds specification	Set FAULT indication (RED)
AVGPWR – Average power from amplifier exceeds specification	Set FAULT indication (RED)
VSWR - Load VSWR exceeds specification	Set FAULT indication (RED)
GATE - UNBLANK gate length exceeds specification	Set FAULT indication (RED)
JUNCTION TEMPERATURE - RF device junction temperature exceeds specification	Set FAULT indication (RED)
UNDEFINED FAULT - All other RF Amplifier faults	Set FAULT indication (RED)
HV RELAY LOW - Power monitor 1 or 2 fault; safety limit allowed for scan parameters has been reached	Set PMON indication (RED)
OVERHEAT - SRFD II Module has exceeded maximum operating temperature	Set FAULT indication (RED)
CABLE FAULT - Bad or disconnected internal cable	Set FAULT indication (RED)
POWER MONITOR 1 FAULT – Combiner dump resistor has exceeded over-temperature limit. Does NOT illuminate PMON LED for power monitor fault conditions.	Set PMON indication (RED)
POWER MONITOR 2 FAULT – Combiner dump resistor has exceeded over-temperature limit. Does NOT illuminate PMON LED for power monitor fault conditions.	Set PMON indication (RED)
OPERATE – Module in operate state	Set OPERATE indication (GREEN)
WAIT - Module in transient state (processing a command)	Set WAIT indication (YELLOW)
STANDBY - Module in standby state	Set STANDBY indication (GREEN)

3-2-1 Error/Status Reporting (Continued)

TABLE 3-1 (Continued)
ERROR/STATUS DEFINITIONS

Error/Status	Condition Front Panel Display
POWER OFF - Module in power off state	Set OFF indication (YELLOW)
HEAD MODE - Module switched to Head mode	Set HEAD indication (GREEN)
BODY MODE - Module switched to Body mode	Set BODY indication (GREEN)
TEST MODE - Module switched to Test mode	Set TEST indication (GREEN)
UNBLANK - Amplifier gating signal received from SSM and in agreement with gating signal from PWR MON	Toggle UNBLANK LED with gate signal ⁽¹⁾ (YELLOW)
HV SUPPLY - High voltage supply for SRFD II is active/not discharged to a safe level	Set HV SUPPLY indication ⁽²⁾ (RED)
POWER ON - AC line power to the SRFD II Module is ON	Set POWER indication (GREEN)

(1) The UNBLANK LED shall be illuminated for a minimum of 100 msec whenever triggered by the gating signal, regardless of the repetition time of the gating pulses. For repetition times less than 100 msec, the UNBLANK LED shall be illuminated continuously.

(2) The HV SUPPLY indication may be an internal visual display rather than a front panel display. However, it shall be clearly visible to the FE when the module cover is removed.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	April 9, 2003	K. Keshena	Initial release.
1	June 5, 2003	D. Thome	Correct error in Table 3-1 concerning PMON LED.